GG-T-671/3B
March 10, 1981
SUPERSEDING
Fed. Spec. GG-T-671/3A
November 24, 1975

FEDERAL SPECIFICATION SHEET

TRIANGLES, DRAFTING, ADJUSTABLE, PLASTIC

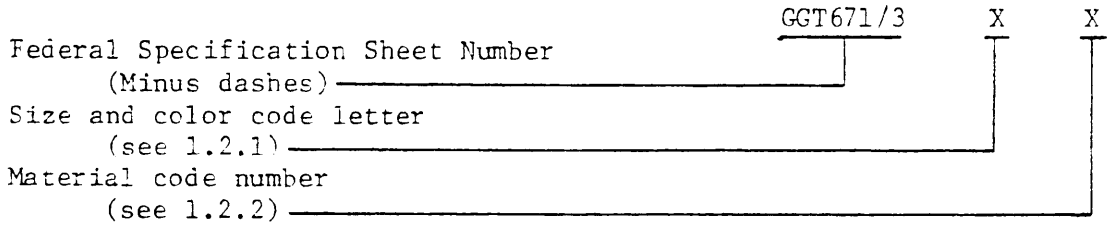
This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

The complete requirements for procuring triangles described herein shall consist of this document and the latest issue in effect of Federal Specification GG-T-671.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification sheet covers the specific requirements for Type III adjustable plastic triangles equipped with protractor elements.

1.2 Classification. The triangles covered by this specification shall have a definitive Federal specification part number and shall be designated in the following form (see 6.2 and 6.3).



FSC 6675

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1.2.1 Size and color of triangles. The size and color of triangles are identified by a single letter (see Table I).

Table I. Size and color code letter

Size 1/ (inches)	Color	
	A	B
6	C	F
8	D	G
10	E	H

Color A - Transparent clear
Color B - Transparent green tinted

Note:
1/ The size of the triangle is the length of the longest side other than the hypotenuse.

1.2.2 Material. The material from which the traingle is made is identified by a one digit (see Table II).

Table II. Material code number

Code number	Material
1	Acrylic
2	Methacrylate
3	Polystyrene
4	Vinyl plastic

3. REQUIREMENTS

3.1 Description. Triangles shall be in accordance with GG-T-671 and as specified herein. Each triangle shall consist of a base angle arm and an adjustable angle arm, a graduated protractor element, and a locking device. With the arms in a closed position, the overall shape shall form a 45 degree right angle triangle, except for a small cutaway portion on the right-hand side of the triangle base as necessary to allow for adjustment movement. The adjustable triangles shall be designed to allow adjustment of the arms to form all angles between the hypotenuse and the base, through a range of from 45 degrees in the closed position to 90 degrees in the open position.

3.2 Triangle arms. The triangle arms shall be plastic, with a minimum thickness of 0.090 inch and a maximum thickness of 0.150 inch.

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3.2.1 Adjustable arm. The adjustable angle arms shall be securely fastened to the protractor element. The part of the adjustable angle arm extending beyond the protractor element shall be shaped to form a 15 degree angle.

3.2.2 Base arm. The base arm shall be pivoted to the protractor element to allow for movement of the adjustable arm. The part of the arm extending beyond the protractor element shall be shaped to form a 30 degree angle. An index mark shall be cut, stamped, etched, indented, or securely fastened on the arm to be used as an index for the graduated arc of the protractor element. The index mark shall be of the same width and legibility as the graduation marks on the protractor.

3.3 Protractor element. The protractor shall be the same material as specified in 3.2. The arc of the protractor element shall have a radius measured from the pivot point of not less than one-half of the size of the adjustable triangle. The arc of the protractor element shall be beveled and graduated in 1/2 degrees. Graduation marks shall be uniform and not greater than 0.010 inch nor less than 0.005 inch width and shall be of three lengths, the shortest length mark denoting 1/2 degree, the medium length mark denoting 1 degree, and the longest mark denoting 5 degree increments. Graduation marks shall be numbered at least every 10 degrees in two rows. The numbers on the outer row shall read clockwise 0 degree to 40 degrees with an additional number at the 45 degree mark. The numbers on the inner row shall read counterclockwise 50 degrees to 90 degrees. The 50 degree mark number shall coincide with the 40 degree mark number, and the 90 degree mark number shall coincide with the 0 degree mark number. Graduation marks and numbers shall be on the upper surface of the protractor element and shall be cut, stamped, etched, or indented. Graduations and numbers shall be accurate and shall be filled with permanent black coloring.

3.4 Locking device. A threaded metal stud and insert with a durable friction washer and knurled knob shall be provided to readily clamp the arms and protractor element in any desired position from 45 degrees to 90 degrees. The knob shall be of suitable size, strength, and shape to be used in lifting and handling the complete triangle.

3.5 Pivoting and fastening devices. Noncorrosive pivoting and fastening devices shall be used. The pivoting device shall make a snug fit, without play, and shall allow free adjustment of the triangle arms.

3.6 Accuracy. The angles shall be within plus or minus 2 minutes of the angles indicated on the protractor. The outside or ruling edges shall be straight and true within plus 0.003 inch. The surfaces of the triangle shall be flat and true within 0.010 inch.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Permanency of graduations test. Moisten a clean white cloth with water and rub it along the graduated area while exerting pressure and keeping the same portion of the cloth in contact with the protractor element at all times. Rub the graduations not less than 10 times in each direction. Any adverse effect on the graduations or any traces of coloring matter on the cloth shall constitute failure of this test.

6. NOTES

6.1 Intended use. The triangles are used in drafting and preparing drawings.

6.2 Ordering data. Procurement documents should specify those desired options offered in GG-T-671 and should also specify the following:

(a) Federal specification sheet part number required (see 1.2).

6.3 Federal specification part number definition. The Federal specification part number is a definitive part number which corresponds to the size and color of triangles covered by this specification sheet and defines the requirements of the options presented under this specification. The Federal specification part number is formed by combining the Federal specification number (without the dashes) with a dash after it, the size and color of triangle code letter, and the material code number as shown in 1.2.

MILITARY INTERESTS:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

GG-T-671/3B Trinagles, Drafting, Adjustable, Plastic

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